EduNova AI — Complete Development Guide

Project

Name: EduNova AI
Tagline: Learn smarter. Solve anything. Discover opportunities.
Capstone Title: EduNova AI: A Multimodal AI-Powered Education Tracker, Multi-LLM Doubt Solver, and Personalized Hackathon Recommendation Platform

EduNova AI helps students:

- Track subjects, attendance, assignments, exams, and study progress.
- Upload PDFs, photos, images, screenshots, handwritten notes, and prompts.
- Ask questions and receive answers from multiple AI models.
- Compare answers from ChatGPT, DeepSeek, Gemini, and Claude.
- Discover live hackathons.
- Get personalized hackathon recommendations with a match score and reason.
- Register for hackathons through official links.

1. Apps and Websites You Will Use

Development Tools

Tool / Website	Purpose	Link
VS Code	Main code editor	https://code.visualstudio.com/
GitHub	Store code, version control, portfolio	https://github.com/
Git	Version control	https://git-scm.com/
Node.js	Required for React/Next.js frontend	https://nodejs.org/
Python	AI backend, RAG, OCR, recommendation system	https://www.python.org/
Docker	Package and deploy backend easily	https://www.docker.com/
Postman	Test APIs	https://www.postman.com/
Vercel	Deploy Next.js frontend	https://vercel.com/
Render / Railway	Deploy Python FastAPI backend	https://render.com/ / https://railway.app/
Hugging Face Spaces	Optional AI demo hosting	https://huggingface.co/spaces

AI Tools for Building

Tool	Purpose
Antigravity	AI coding/agent assistance for building the website
Manus	AI agent for generating UI, code, documentation, and workflows
ChatGPT	AI answers, coding help, documentation
Claude	AI answers and high-quality writing/code help
Gemini	Multimodal AI for text, image, PDF, and code
DeepSeek	Low-cost AI model for chat and reasoning
v0	Generate React/Tailwind UI components
Figma	Design website screens before coding
Framer Motion	Add smooth animations
Lucide	Modern icons

Backend and Database

Tool	Purpose	Link
Supabase	Authentication, PostgreSQL database, file storage	https://supabase.com/
Supabase Auth	Student login/signup	https://supabase.com/docs/guides/auth
Supabase Storage	Store uploaded PDFs and images	https://supabase.com/docs/guides/storage
FastAPI	Python AI backend	https://fastapi.tiangolo.com/
Uvicorn	Run FastAPI server	https://www.uvicorn.org/
FAISS	Vector search for PDF RAG	https://github.com/facebookresearch/faiss
Chroma	Easy vector database	https://www.trychroma.com/
LangChain	RAG and LLM workflow	https://www.langchain.com/
LlamaIndex	Alternative RAG framework	https://www.llamaindex.ai/

AI Provider Documentation

Provider	Documentation
OpenAI	https://platform.openai.com/docs
DeepSeek	https://api-docs.deepseek.com/
Google Gemini	https://ai.google.dev/
Anthropic Claude	https://docs.anthropic.com/

Hackathon Data Sources

Source	Use
Devpost	Global and AI hackathons
Unstop	Indian student hackathons and competitions
MLH	Student-focused hackathons
Devfolio	Indian developer hackathons
HackerEarth	Corporate and skill challenges
Hack2Skill	Hackathons and innovation challenges

2. Recommended Website Stack

Frontend

- **Next.js + React**
- **Tailwind CSS**
- **shadcn/ui**
- **Framer Motion**
- **Lucide React**
- **Recharts**
- **React Hook Form + Zod**

Backend

- **Python FastAPI**
- **Uvicorn**
- **Supabase Python client**
- **LangChain**
- **FAISS or Chroma**
- **PyMuPDF / pdfplumber**
- **Pytesseract**
- **APScheduler**

Database and Storage

- **Supabase PostgreSQL**
- **Supabase Auth**
- **Supabase Storage**

AI Providers

- OpenAI
- DeepSeek
- Gemini
- Anthropic Claude

3. Website Pages

Public Pages

1. **Landing Page**
    - Hero section
    - Features
    - How it works
    - Hackathon preview
    - Pricing optional
    - Footer
2. **Login / Signup**
    - Email/password
    - Google login optional
    - Supabase Auth

Student Pages

3. **Dashboard**
    - Study streak
    - Attendance
    - Assignments due
    - Exam countdown
    - Subject progress
    - AI recommendations
4. **Subjects**
    - Add/edit/delete subjects
    - Add chapters and topics
    - Mark topics complete
    - Track progress
5. **Study Materials**

    - Upload PDF, image, photo, screenshot, notes
    - View uploaded files
    - Search materials
    - Delete files
6. **AI Doubt Solver**
    - Select uploaded material
    - Ask a question
    - Choose AI provider
    - Get answer with source reference
7. **Multi-AI Compare**
    - Ask one question
    - Select multiple AI models
    - Compare answers side by side
    - Save the best answer
8. **Quiz Generator**
    - Generate MCQs
    - Generate short answers
    - Generate flashcards
    - Generate revision notes
9. **Hackathons**
    - Live hackathon feed
    - Search and filters
    - Deadline countdown
    - Match score
    - Why recommended
    - Register Now button
10. **Profile**

- Name, college, branch
- Skills
- Interests
- Preferred hackathon mode
- Resume optional

4. Color Palette and UI Design

Use a dark futuristic AI dashboard theme.

Use	Color	Hex
Main background	Deep navy	`#0B1020`
Card background	Dark slate	`#151B2B`
Secondary background	Slate	`#111827`
Primary	Electric indigo	`#6366F1`
Secondary	Violet	`#8B5CF6`
Accent	Cyan	`#22D3EE`
Text	Soft white	`#E5E7EB`
Muted text	Gray	`#94A3B8`
Border	Transparent white	`rgba(255,255,255,0.08)`
Success	Green	`#22C55E`
Warning	Amber	`#F59E0B`
Error	Rose	`#F43F5E`

UI Rules

- Use rounded corners: 16px.
- Use glassmorphism cards with subtle blur.
- Use indigo for primary buttons.
- Use cyan for AI-related actions.
- Use violet only in gradients.
- Keep animations subtle and fast.
- Make the site mobile-first.
- Use Inter or Plus Jakarta Sans for text.
- Use Space Grotesk for headings.
- Use JetBrains Mono for code.

Smooth Scrolling CSS

```
@media (prefers-reduced-motion: no-preference) {
  html {
    scroll-behavior: smooth;
  }
}
```

Framer Motion Example

```
<motion.div
  initial={{ opacity: 0, y: 24 }}
  whileInView={{ opacity: 1, y: 0 }}
  viewport={{ once: true }}
  transition={{ duration: 0.6, ease: "easeOut" }}
>
  <HackathonCard />
</motion.div>
```

5. Complete Procedure

Step 1: Plan the Project

Create these documents:

- Project abstract
- Problem statement
- Objectives
- Features list
- System architecture
- Database design
- API design
- UI wireframes
- Timeline

Step 2: Create GitHub Repository

1. Create a GitHub account.
2. Create a new repository named `edunova-ai`.
3. Add a README.
4. Clone it locally:

```
git clone https://github.com/yourusername/edunova-ai.git
cd edunova-ai
```

Step 3: Create Frontend

Use Next.js:

```
npx create-next-app@latest frontend
cd frontend
```

```
npm install
```

Install UI and animation packages:

```
npm install tailwindcss framer-motion lucide-react recharts react-hook-form zod @supa
```

Initialize shadcn/ui:

```
npx shadcn@latest init
```

Step 4: Set Up Supabase

1. Go to https://supabase.com/.
2. Create a new project.
3. Copy the project URL and anon key.
4. Enable Email Authentication.
5. Create tables in Supabase SQL Editor.
6. Create a storage bucket named `study-materials`.
7. Add environment variables to `.env.local`:

```
NEXT_PUBLIC_SUPABASE_URL=your_supabase_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_supabase_anon_key
```

Step 5: Build Authentication

Use Supabase Auth for:

- Student signup
- Login
- Logout
- Protected routes
- Profile creation

Step 6: Build Education Tracker

Create pages for:

- Dashboard
- Subjects
- Assignments
- Exams

- Attendance
- Study sessions

Use Supabase tables:

- users
- subjects
- topics
- assignments
- exams
- attendance
- study_sessions

Step 7: Build AI Backend

Create a separate Python folder:

```
mkdir backend
cd backend
python -m venv venv
source venv/bin/activate  # Windows: venv\Scripts\activate
pip install fastapi uvicorn python-dotenv supabase langchain faiss-cpu pymupdf pytes
```

Create `main.py` for FastAPI.

Create API routes:

- `/upload`
- `/ask`
- `/compare`
- `/quiz`
- `/hackathons`
- `/recommendations`

Step 8: Add PDF RAG

Workflow:

1. Student uploads PDF.
2. FastAPI extracts text.
3. Text is split into chunks.
4. Chunks are converted into embeddings.
5. Embeddings are stored in FAISS/Chroma.

6. When student asks a question, relevant chunks are retrieved.
7. Retrieved chunks are sent to the selected AI model.
8. AI returns an answer with references.

Step 9: Add Image and Photo Support

Use:

- Pytesseract for printed/handwritten text.
- Gemini or GPT vision model for diagrams, charts, screenshots, and handwritten questions.

Workflow:

1. Student uploads image.
2. OCR extracts text if available.
3. Vision model describes diagrams or visual content.
4. Extracted content is combined.
5. AI model answers the student question.

Step 10: Add Multi-LLM Comparison

Create provider modules:

- `openai_provider.py`
- `deepseek_provider.py`
- `gemini_provider.py`
- `anthropic_provider.py`

Each provider receives:

- Student question
- Retrieved PDF/image context
- Selected model

Then return:

- Answer
- Response time
- Source references

Step 11: Add Hackathon Discovery

Create a scheduled job using APScheduler.

Fetch data from:

- Devpost API

- Unstop API
- Other approved sources

Normalize every hackathon into:

```
{
  "title": "",
  "description": "",
  "source": "",
  "registration_url": "",
  "mode": "",
  "location": "",
  "deadline": "",
  "prize": "",
  "tags": []
}
```

Store it in Supabase.

Step 12: Add Hackathon Recommendation

Create a student profile with:

- Skills
- Interests
- Experience level
- Preferred mode
- Location
- Saved hackathons

Calculate match score:

```
Match Score =
(0.35 × Skill Match) +
(0.25 × Interest Match) +
(0.15 × Deadline Urgency) +
(0.10 × Mode/Location Match) +
(0.10 × Experience Match) +
(0.05 × Past Behavior)
```

Use TF-IDF and cosine similarity for skill/theme matching.

Show:

- Match score
- Why recommended
- Deadline countdown
- Register Now button

Step 13: Add Animations

Use Framer Motion for:

- Fade-up animations
- Card hover effects
- Page transitions
- AI typing effect
- Progress-bar animation
- Hackathon card entrance

Step 14: Test the Website

Test:

- Signup/login
- Subject creation
- File upload
- PDF question answering
- Image question answering
- Multi-AI comparison
- Hackathon fetching
- Hackathon recommendation
- Mobile responsiveness
- Error handling
- API failures

Step 15: Deploy

- Deploy frontend to Vercel.
- Deploy FastAPI backend to Render or Railway.
- Host Supabase database in Supabase Cloud.
- Store API keys in environment variables.
- Connect frontend to backend using a public API URL.

6. GitHub Projects to Study

PDF RAG Projects

1. **PDF AI Assistant**
   https://github.com/Yash6606/pdf-ai-assistant
   Study: RAG, OCR, FAISS, BM25, reranking, LLM Q&A.
2. **Multimodal RAG Knowledge Assistant**
   https://github.com/loid122/Multimodal-RAG-Knowledge-Assistant
   Study: PDF text, tables, images, embeddings, summaries.
3. **Multimodal RAG Assistant**
   https://github.com/amiraa23/multimodal-rag-assistant
   Study: Streamlit, PDF/CSV analysis, image captioning, Whisper, RAG.
4. **Multimodal RAG using Gemini**
   https://github.com/MukulLambat/multimodal_RAG
   Study: Gemini, text/table/image processing, vector search.
5. **PDF/Image OCR Question Answering**
   https://github.com/crystalkhem/PDF-Image-Text-Extraction-Question-Answering-OCR-LLM/
   Study: OCR, PDF processing, image question answering.

Education AI Projects

6. **AI Education Assistant**
   https://github.com/omarabushah/AI-education-assistant
   Study: Streamlit education dashboard, student/teacher roles, GPT integration.
7. **Multi-Agent Study Assistant**
   https://github.com/A-R007/Multi-Agent-Study-Assistant
   Study: personalized learning, quizzes, tutoring, RAG document Q&A.

Hackathon / Recommendation Projects

8. **CollaborateAI**
   https://github.com/aid278114/CollaborateAI
   Study: hackathon team formation and hybrid recommendations.
9. **ASSEMBLE — AI Skill Matching Platform**
   https://github.com/mounika4785/ASSEMBLE-AI-Skill-Matching-Platform
   Study: matching students with hackathons, projects, teams, and opportunities.
10. **Event Recommendation System**
    https://github.com/thiranaas/Event_Recommendation_System
    Study: event discovery using skills, interests, event type, and mode.

7. Important GitHub Links

Resource	Link
Next.js	https://github.com/vercel/next.js
Tailwind CSS	https://github.com/tailwindlabs/tailwindcss
shadcn/ui	https://github.com/shadcn-ui/ui
Framer Motion	https://github.com/motiondivision/motion
Supabase	https://github.com/supabase/supabase
FastAPI	https://github.com/fastapi/fastapi
LangChain	https://github.com/langchain-ai/langchain
FAISS	https://github.com/facebookresearch/faiss
Chroma	https://github.com/chroma-core/chroma
Tesseract OCR	https://github.com/tesseract-ocr/tesseract
OpenAI Python SDK	https://github.com/openai/openai-python
Gemini Python SDK	https://github.com/googleapis/python-genai
Devpost	https://github.com/topics/devpost

8. Folder Structure

```
edunova-ai/
├── frontend/
│   ├── app/
│   │   ├── page.tsx
│   │   ├── dashboard/page.tsx
│   │   ├── subjects/page.tsx
│   │   ├── materials/page.tsx
│   │   ├── ai-solver/page.tsx
│   │   ├── compare/page.tsx
│   │   ├── hackathons/page.tsx
│   │   └── profile/page.tsx
│   ├── components/
│   │   ├── Navbar.tsx
│   │   ├── Sidebar.tsx
│   │   ├── SubjectCard.tsx
│   │   ├── AIAnswerCard.tsx
│   │   ├── HackathonCard.tsx
│   │   └── UploadBox.tsx
│   ├── lib/
│   │   ├── supabase.ts
│   │   └── api.ts
│   └── package.json
│
├── backend/
│   ├── main.py
│   ├── routers/
```

```
│   │   ├── auth.py
│   │   ├── documents.py
│   │   ├── ai.py
│   │   └── hackathons.py
│   ├── services/
│   │   ├── pdf_service.py
│   │   ├── ocr_service.py
│   │   ├── rag_service.py
│   │   ├── ai_router.py
│   │   └── recommendation_service.py
│   ├── providers/
│   │   ├── openai_provider.py
│   │   ├── deepseek_provider.py
│   │   ├── gemini_provider.py
│   │   └── anthropic_provider.py
│   ├── requirements.txt
│   └── .env
│
├── docs/
│   ├── project-report.md
│   ├── architecture.md
│   └── api-documentation.md
│
└── README.md
```

9. Environment Variables

Frontend `.env.local`

```
NEXT_PUBLIC_SUPABASE_URL=your_supabase_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_supabase_anon_key
NEXT_PUBLIC_API_URL=http://localhost:8000
```

Backend `.env`

```
SUPABASE_URL=your_supabase_url
SUPABASE_SERVICE_ROLE_KEY=your_service_role_key
OPENAI_API_KEY=your_openai_key
DEEPSEEK_API_KEY=your_deepseek_key
GEMINI_API_KEY=your_gemini_key
ANTHROPIC_API_KEY=your_anthropic_key
```

Never upload `.env` files to GitHub.

10. Development Timeline

Week	Work
Week 1	Finalize requirements, UI design, database schema
Week 2	Build frontend layout, routing, Supabase auth
Week 3	Build education tracker and dashboard
Week 4	Build PDF upload and text extraction
Week 5	Build RAG question answering
Week 6	Add image OCR and vision AI
Week 7	Add multi-LLM comparison
Week 8	Build hackathon collector and database
Week 9	Build recommendation system
Week 10	Add animations, polish UI, responsive design
Week 11	Testing, bug fixing, deployment
Week 12	Documentation, demo video, capstone presentation

11. Capstone Presentation Tips

Show these during your demo:

1. Student login.
2. Add subjects and attendance.
3. Upload a PDF.
4. Ask a question and show a grounded answer.
5. Upload a handwritten photo and solve it.
6. Compare ChatGPT, DeepSeek, Gemini, and Claude answers.
7. Open Hackathons page.
8. Show a recommended hackathon and explain the match score.
9. Click Register Now to open the official page.
10. Show dashboard analytics and exam-readiness prediction.

12. Final Advice

- Build the frontend and Supabase auth first.
- Then build the education tracker.
- Then add one AI provider.
- Then add RAG for PDFs.

- Then add image/OCR support.
- Then add multi-AI comparison.
- Then add hackathon discovery and recommendations.
- Do not try to build every feature on day one.

Your unique strength is the combination of:

Education Tracker + Multimodal AI Doubt Solver + Multi-LLM Comparison + Live Hackathon Recommendations